

FRANCK DEBA

Quantitative Researcher — Data Scientist — Financial Engineering
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Education

Master M2MO (Formerly DEA Laure Elie)

Quantitative Finance and Stochastic Modelling

September 2025 – Present

Université Paris Cité, France

WorldQuant University

MSc Degree in Financial Engineering

October 2024 – Present

Online

ENSAI

Engineer's Degree in Data Science and Risk Management

August 2023 – December 2025

Rennes, France

Experience

Quantitative Risk Analyst Intern | Crédit Agricole

April 2025 – September 2025

Risk Modeling Department

Paris, France

- Developed climate stress-testing methodology using linear regression and ECM modeling for risk assessment
- Engineered automated BackTesting tool in Python (Dash) for Probability of Default model validation
- Migrated statistical models from SAS to cloud-based Python platform

Quantitative Modeling Intern | Groupe VYV - Harmonie Mutuelle

June 2024 – August 2024

Assurance Prévoyance

Paris, France

- Applied unsupervised learning (K-Means clustering) to segment insured populations for mental health risk
- Quantified risk exposure by implementing survival analysis models (Cox Proportional Hazards)
- Modeled and forecasted treatment costs using generalized linear models (GLMs)

Quantitative Research Projects

Yield Curve Modeling | Python

January 2025

- Constructed zero-coupon yield curves from market data using bootstrapping methodology
- Calibrated Hull-White model for interest rate derivatives pricing and risk management
- Implemented vanilla caplet pricing using Black model and analyzed volatility smile patterns
- Developed Monte Carlo simulations for pricing Bermudan barrier interest rate options

Algorithmic Trading Strategy | Python

December 2024

- Modeled stock returns using VAR with FARIMA-GARCH volatility processes
- Implemented LSTM and Prophet models for multi-horizon price forecasting
- Built interactive Dash platform for strategy backtesting and visualization

Multi-Asset Risk Management | Python

March 2025

- Calculated Credit Value-at-Risk (CreditVaR) for bond portfolios using Monte Carlo simulation
- Modeled cross-asset dependencies using Copula functions (Gaussian, t-Copula) for tail risk assessment
- Implemented stress testing scenarios for portfolio risk under extreme market conditions

Technical Skills

Quantitative Finance

Modeling: Stochastic Calculus, Options Pricing, Risk Metrics (VaR, ES), Portfolio Optimization

Methods: Monte Carlo Simulation, Time Series Analysis, Yield Curve Bootstrapping, Copulas

Domains: Market Risk, Credit Risk, Climate Risk, Derivatives Pricing

Programming & Tools

Languages: Python (Advanced), C++, R, SQL, SAS

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Statsmodels

Tools: Git, Linux, AWS, Power BI, LaTeX, Dash

Languages & Certifications

French: Native

English: Professional Proficiency (TOEFL/TOEIC Certified)

Certifications: AMF, AI in Finance (World Bank), Financial Markets (Yale)